

C-DAC — ASSIGNMENT 02 ANSWERS

QUESTION 01

Electricity Bill Calculation — if-else-if. Read units in main(). Pass to static method calculateBill(). (Slabs: 0-100: ■2, 101-200: ■3, 201-300: ■5, Above 300: ■7).

LOGIC

Read units in `main()`. Pass `units` to static method `calculateBill()`. Use slab logic to sum tier costs and return the bill amount.

CODE

```
// Assignment 02 - Question 01: Electricity Bill Calculation
public class Q01 {
    static double calculateBill(int units) {
        double bill = 0.0;
        if (units <= 100) {
            bill = units * 2.0;
        } else if (units <= 200) {
            bill = (100 * 2.0) + ((units - 100) * 3.0);
        } else if (units <= 300) {
            bill = (100 * 2.0) + (100 * 3.0) + ((units - 200) * 5.0);
        } else {
            bill = (100 * 2.0) + (100 * 3.0) + (100 * 5.0) + ((units - 300) * 7.0);
        }
        return bill;
    }

    public static void main(String[] args) {
        int units = 250;
        double totalBill = calculateBill(units);
        System.out.println("Units Consumed: " + units);
        System.out.println("Total Electricity Bill: Rs. " + totalBill);
    }
}
```

EXPECTED OUTPUT

```
Units Consumed: 250
Total Electricity Bill: Rs. 750.0
```

QUESTION 02

Student Grade Calculation — if-else-if. Read marks in main(). Pass to static method calculateGrade(). (90-100: A, 75-89: B, 60-74: C, 50-59: D, Below 50: F).

LOGIC

Pass marks to static method `calculateGrade()`. Evaluate slab ranges using `if-else-if` ladder and return the corresponding grade char.

CODE

```
// Assignment 02 - Question 02: Student Grade Calculation
public class Q02 {
    static char calculateGrade(int marks) {
        if (marks >= 90 && marks <= 100) {
            return 'A';
        } else if (marks >= 75) {
            return 'B';
        } else if (marks >= 60) {
            return 'C';
        } else if (marks >= 50) {
            return 'D';
        } else {
            return 'F';
        }
    }

    public static void main(String[] args) {
        int marks = 82;
        char grade = calculateGrade(marks);
        System.out.println("Student Marks: " + marks);
        System.out.println("Calculated Grade: " + grade);
    }
}
```

EXPECTED OUTPUT

```
Student Marks: 82
Calculated Grade: B
```

QUESTION 03

Product Discount — if-else-if. Read productId, productName, price in main(). Pass price to static method calculateFinalPrice().

LOGIC

Pass price to static method `calculateFinalPrice()`. Calculate discount percentage based on price range and return discounted price.

CODE

```
// Assignment 02 - Question 03: Product Discount
public class Q03 {
    static double calculateFinalPrice(double price) {
        double discount = 0.0;
        if (price >= 10000) {
            discount = price * 0.20;
        } else if (price >= 5000) {
            discount = price * 0.10;
        } else if (price >= 2000) {
            discount = price * 0.05;
        } else {
            discount = 0.0;
        }
        return price - discount;
    }

    public static void main(String[] args) {
        int productId = 101;
        String productName = "Laptop";
        double price = 12000.0;

        double finalPrice = calculateFinalPrice(price);

        System.out.println("Product ID: " + productId);
        System.out.println("Product Name: " + productName);
        System.out.println("Original Price: Rs. " + price);
        System.out.println("Final Price after Discount: Rs. " + finalPrice);
    }
}
```

EXPECTED OUTPUT

```
Product ID: 101
Product Name: Laptop
Original Price: Rs. 12000.0
Final Price after Discount: Rs. 9600.0
```

QUESTION 04

Largest of Two Numbers — if-else. Read two integers in main(). Pass to static method findLargest().

LOGIC

Pass two numbers to static method `findLargest()`. Compare using `a >= b` and return larger value.

CODE

```
// Assignment 02 - Question 04: Largest of Two Numbers
public class Q04 {
    static int findLargest(int a, int b) {
        if (a >= b) {
            return a;
        } else {
            return b;
        }
    }

    public static void main(String[] args) {
        int num1 = 45;
        int num2 = 89;

        int largest = findLargest(num1, num2);
        System.out.println("Number 1: " + num1);
        System.out.println("Number 2: " + num2);
        System.out.println("Largest Number: " + largest);
    }
}
```

EXPECTED OUTPUT

```
Number 1: 45
Number 2: 89
Largest Number: 89
```

QUESTION 05

Check Voting Eligibility — if. Read age in main(). Pass to static method isEligible() returning boolean.

LOGIC

Pass age to static method `isEligible()`. Return `true` if `age >= 18`, else `false`.

CODE

```
// Assignment 02 - Question 05: Check Voting Eligibility
public class Q05 {
    static boolean isEligible(int age) {
        if (age >= 18) {
            return true;
        } else {
            return false;
        }
    }

    public static void main(String[] args) {
        int age = 21;
        boolean eligible = isEligible(age);

        System.out.println("Age: " + age);
        if (eligible) {
            System.out.println("Status: Eligible to vote.");
        } else {
            System.out.println("Status: Not eligible to vote.");
        }
    }
}
```

EXPECTED OUTPUT

```
Age: 21
Status: Eligible to vote.
```

QUESTION 06

Employee Salary Calculation — if-else. Read basic salary in main(). If $\geq$ 50,000 add 20% HRA, else 10% HRA via static method calculateSalary().

LOGIC

Pass basic salary to static method `calculateSalary()`. If `basicSalary  $\geq$  50000`, add 20% HRA, else 10% HRA, and return total salary.

CODE

```
// Assignment 02 - Question 06: Employee Salary Calculation
public class Q06 {
    static double calculateSalary(double basicSalary) {
        double hra;
        if (basicSalary >= 50000) {
            hra = basicSalary * 0.20;
        } else {
            hra = basicSalary * 0.10;
        }
        return basicSalary + hra;
    }

    public static void main(String[] args) {
        double basicSalary = 55000.0;
        double finalSalary = calculateSalary(basicSalary);

        System.out.println("Basic Salary: Rs. " + basicSalary);
        System.out.println("Final Salary (with HRA): Rs. " + finalSalary);
    }
}
```

EXPECTED OUTPUT

```
Basic Salary: Rs. 55000.0
Final Salary (with HRA): Rs. 66000.0
```

QUESTION 07

Temperature Conversion — Operators + if. Convert Celsius to Fahrenheit via static method `convertTemperature()`. In `main()`, check if $> 100^{\circ}\text{F}$.

LOGIC

Pass Celsius to static method `convertTemperature()`. Calculate Fahrenheit and check in `main()` if temperature $> 100^{\circ}\text{F}$.

CODE

```
// Assignment 02 - Question 07: Temperature Conversion
public class Q07 {
    static double convertTemperature(double celsius) {
        return (celsius * 9.0 / 5.0) + 32.0;
    }

    public static void main(String[] args) {
        double celsius = 38.5;
        double fahrenheit = convertTemperature(celsius);

        System.out.println("Celsius Temperature: " + celsius + " deg C");
        System.out.println("Fahrenheit Temperature: " + fahrenheit + " deg F");

        if (fahrenheit > 100) {
            System.out.println("Warning: Temperature is above 100 deg F!");
        } else {
            System.out.println("Temperature is within normal limits.");
        }
    }
}
```

EXPECTED OUTPUT

```
Celsius Temperature: 38.5 deg C
Fahrenheit Temperature: 101.3 deg F
Warning: Temperature is above 100 deg F!
```

QUESTION 08

Simple Calculator — if-else-if. Read numbers and operator (+, -, *, /) in main(). Pass to static method calculate(). Handle division by zero.

LOGIC

Pass two numbers and operator char to static method `calculate()`. Use `switch` block and handle division by zero condition.

CODE

```
// Assignment 02 - Question 08: Simple Calculator
public class Q08 {
    static double calculate(double a, double b, char op) {
        switch (op) {
            case '+': return a + b;
            case '-': return a - b;
            case '*': return a * b;
            case '/':
                if (b != 0) return a / b;
                else {
                    System.out.println("Error: Division by zero.");
                    return Double.NaN;
                }
            default:
                System.out.println("Invalid Operator!");
                return Double.NaN;
        }
    }

    public static void main(String[] args) {
        double num1 = 25.0;
        double num2 = 5.0;
        char operator = '*';

        double result = calculate(num1, num2, operator);
        System.out.println("Num 1: " + num1 + ", Num 2: " + num2 + ", Operator: " + operator);
        System.out.println("Result: " + result);
    }
}
```

EXPECTED OUTPUT

```
Num 1: 25.0, Num 2: 5.0, Operator: *
Result: 125.0
```

QUESTION 09

Bus Ticket Fare — if-else-if. Read name & age in main(). Pass age to static method calculateFare().

LOGIC

Pass age to static method `calculateFare()`. Apply age bracket pricing rules (Below 5 free, 5-12 ■20, 13-59 ■40, 60+ ■25).

CODE

```
// Assignment 02 - Question 09: Bus Ticket Fare
public class Q09 {
    static double calculateFare(int age) {
        if (age < 5) {
            return 0.0;
        } else if (age <= 12) {
            return 20.0;
        } else if (age <= 59) {
            return 40.0;
        } else {
            return 25.0;
        }
    }

    public static void main(String[] args) {
        String passengerName = "Ramesh Kumar";
        int age = 65;

        double fare = calculateFare(age);

        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Passenger Age: " + age);
        System.out.println("Bus Ticket Fare: Rs. " + fare);
    }
}
```

EXPECTED OUTPUT

```
Passenger Name: Ramesh Kumar
Passenger Age: 65
Bus Ticket Fare: Rs. 25.0
```

QUESTION 10

Mobile Data Usage — if-else-if. Read mobile number and GB usage in main(). Pass usage to static method calculateCharge().

LOGIC

Pass usage in GB to static method `calculateCharge()`. Apply data tier rates (Up to 1GB ■50, 5GB ■100, 10GB ■200, >10GB ■350).

CODE

```
// Assignment 02 - Question 10: Mobile Data Usage
public class Q10 {
    static double calculateCharge(double usageGB) {
        if (usageGB <= 1.0) {
            return 50.0;
        } else if (usageGB <= 5.0) {
            return 100.0;
        } else if (usageGB <= 10.0) {
            return 200.0;
        } else {
            return 350.0;
        }
    }

    public static void main(String[] args) {
        long mobileNumber = 9876543210L;
        double dataUsage = 7.5;

        double charge = calculateCharge(dataUsage);

        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("Data Usage: " + dataUsage + " GB");
        System.out.println("Total Data Charge: Rs. " + charge);
    }
}
```

EXPECTED OUTPUT

```
Mobile Number: 9876543210
Data Usage: 7.5 GB
Total Data Charge: Rs. 200.0
```
